

Closing the Pre-Infusion Gap: A Quality Improvement Initiative to Enhance Laboratory Readiness for Outpatient IV Iron Referrals

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Laboratory readiness increased from 69.1% to 89.8%

MOTIVATION

At Renown in Northern Nevada, some patients arrived for IV iron infusions without valid iron studies, requiring same-day testing and coordination among nursing staff, pharmacy, and referring clinicians. Lab orders embedded in treatment plans did not reliably ensure readiness before arrival, highlighting the need for earlier verification.

★
Reno

Nearly 1 in 3 lacked valid iron studies before infusion
Baseline: 64/207 | Jan–Apr 2026

AIM

Increase required iron study completion to ≥85% within six months among internal IV iron referrals, supported by verification before scheduling.

ESTIMATED CHAIR-TIME BURDEN

Baseline period: Jan–Apr 2026
Assuming 75 additional chair-minutes per visit lacking valid iron studies

- 20 excess chair-hours/month
- 240 excess chair-hours/year (annualized)

**Based on an infusion pharmacy staff estimate of 75 extra minutes per affected visit. Annual projection assumes unchanged baseline volume and frequency of missing or invalid studies; delays were not directly measured.*

METHODS

Single-center QI evaluation of 416 internal outpatient IV iron referral records, January 1–September 14, 2026. All records were included regardless of treatment status.

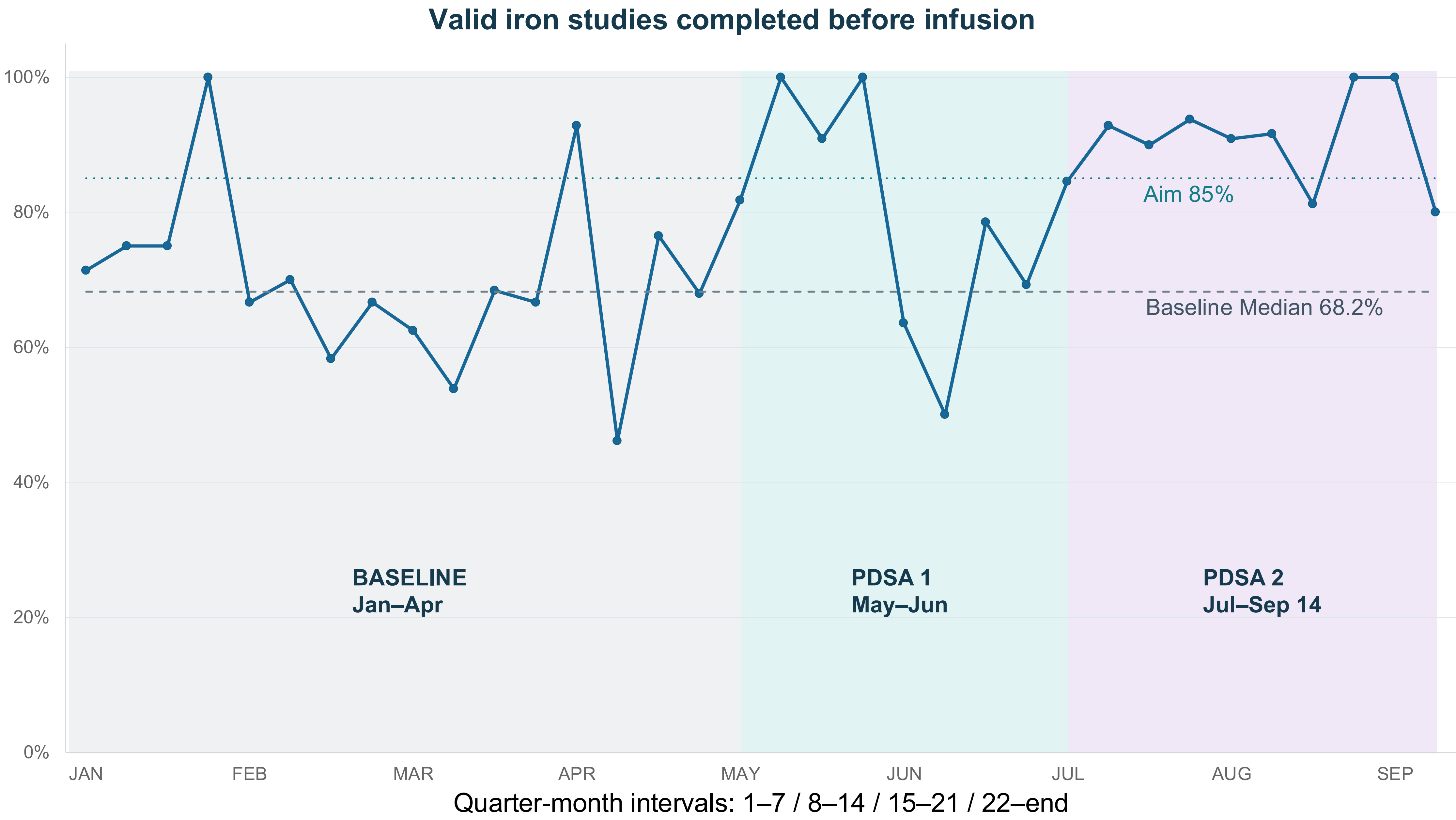
Process measure: Percentage with complete, valid iron studies before infusion.

Outcome measure: Percentage meeting laboratory treatment parameters.

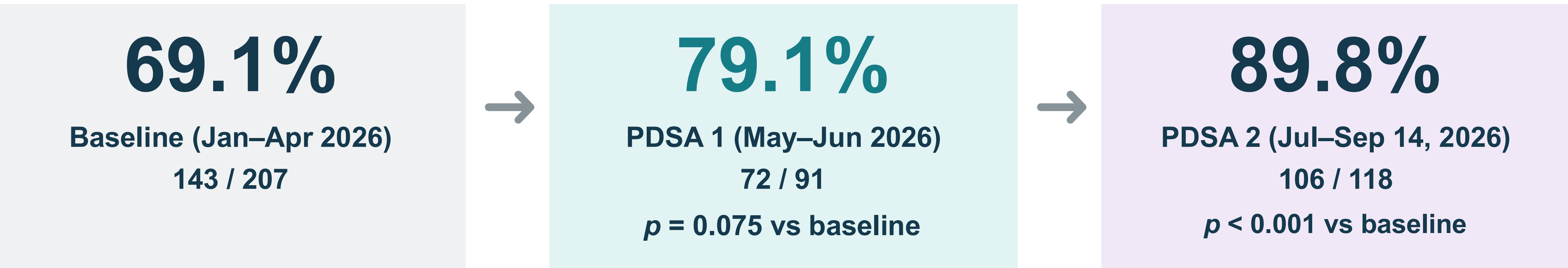
Analysis:

- Quarter-month run chart displayed process performance over time using a fixed baseline median.
- Pearson’s chi-square test compared pooled laboratory readiness rates across the three study periods: baseline, PDSA 1, and PDSA 2, assuming independent records. Outcome rates were summarized descriptively.

PROCESS MEASURE: LABORATORY READINESS OVER TIME

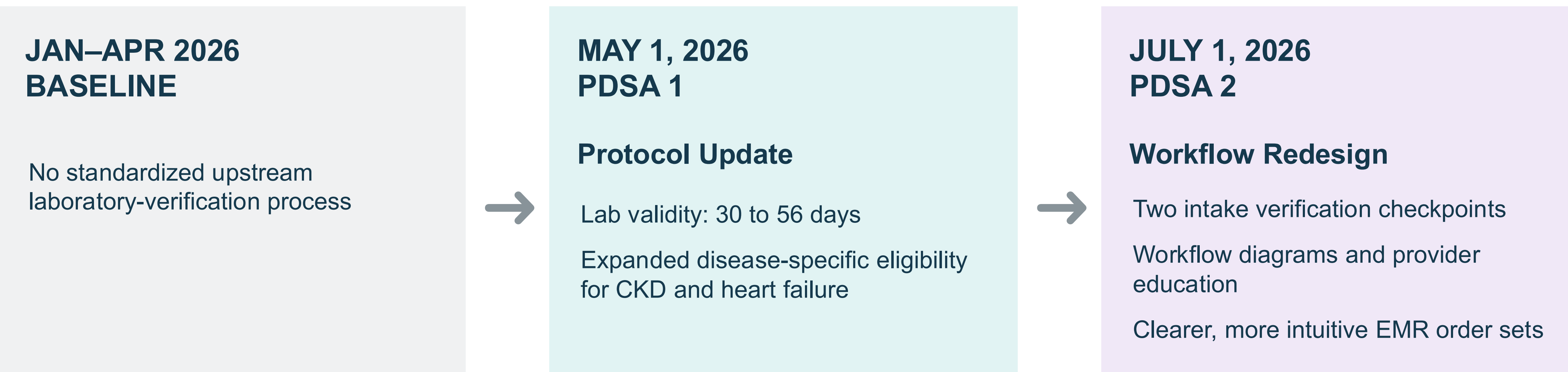


LABORATORY READINESS BY PERIOD



Overall $p < 0.001$ across three periods.
Pairwise p-values compare each intervention period with baseline using Pearson’s chi-square tests with Holm adjustment.

WHAT WE CHANGED



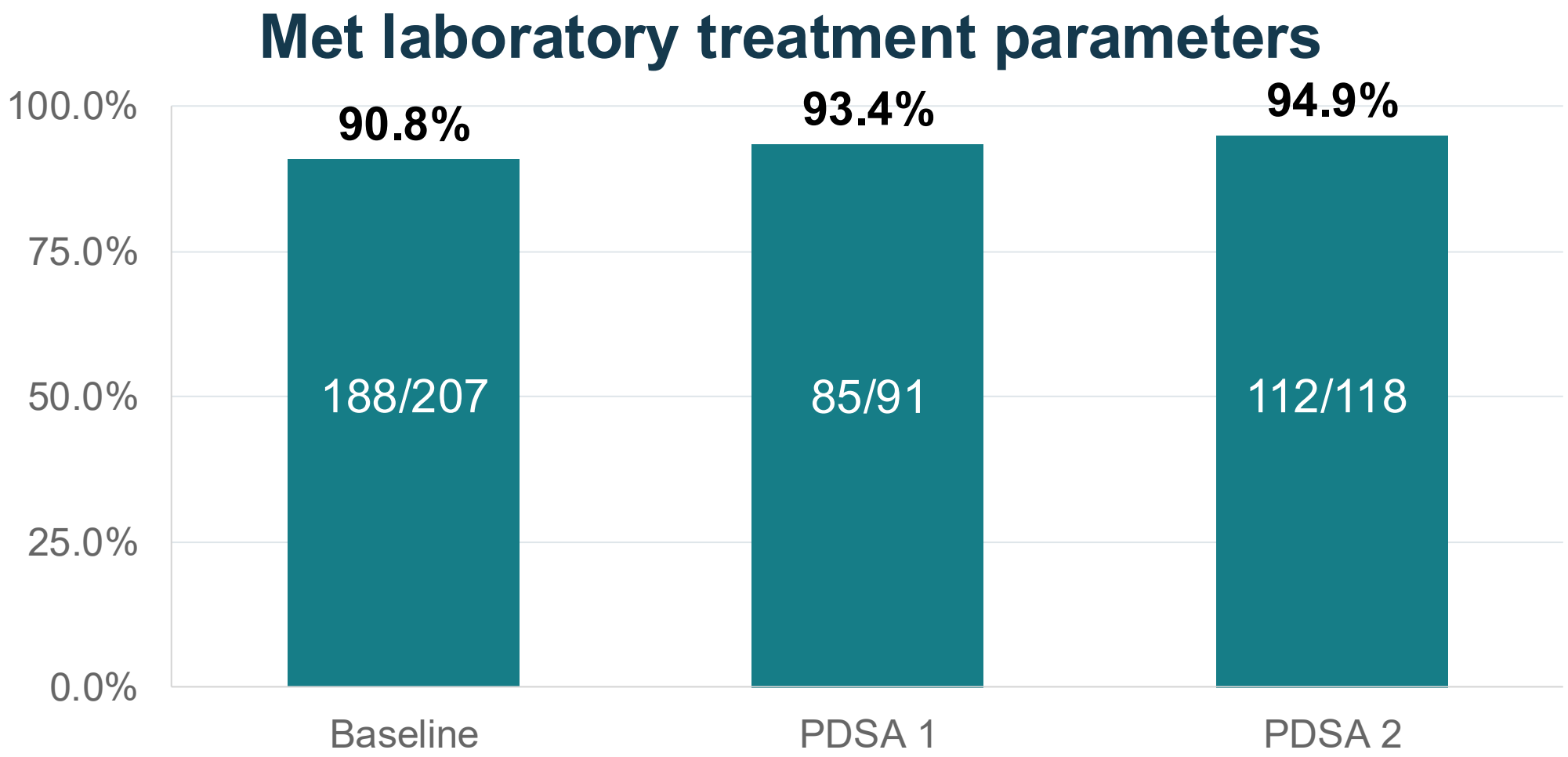
ACKNOWLEDGMENTS

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REFERENCES

Renown Health. *Pricing Transparency*. Accessed September 30, 2026. <https://www.renown.org/Patients-and-Visitors/Billing/Pricing-Transparency>

OUTCOME MEASURE



Attainment remained high across all periods.

PATIENT IMPACT (ESTIMATED)

≈161 patient-hours of extra chair time potentially avoided annually* from baseline to PDSA 2.

Potentially less waiting and fewer treatment disruptions for patients and caregivers, including those traveling from rural Northern Nevada.

MODELED ANNUAL CAPACITY BURDEN

Annual burden*	Baseline	PDSA 1	PDSA 2
Excess chair-hours	240	162	79
Illustrative opportunity value†			
Medicaid	\$14,700	\$9,900	\$4,800
Commercial mean	\$168,100	\$113,500	\$55,300
Hometown Health National	\$279,000	\$188,000	\$92,000

Released chair time could expand access to oncology and other infusion therapies.

** Assumes 621 annual visits, 75 extra chair-minutes per affected visit, and sustained period-specific rates; delay duration was not directly measured.*

† Illustrative gross revenue proxy assuming one additional CPT 96365-eligible visit per released chair-hour and full utilization; not measured costs, lost revenue, or savings.

LIMITATIONS

Single-center, uncontrolled evaluation with limited follow-up. Changes in laboratory validity and eligibility criteria may affect comparisons across periods. Individual intervention effects cannot be isolated. Chair-time and financial estimates are modeled, not directly measured.

CONCLUSION

Protocol and workflow changes were associated with improved laboratory readiness, exceeding the 85% target overall after PDSA 2. Laboratory treatment-parameter attainment remained high. Intermittent below-target intervals highlight the need for continued monitoring and refinement.

NEXT STEPS

Monitor sustainability and adapt the workflow for an external-referral pilot.

